PNTHR

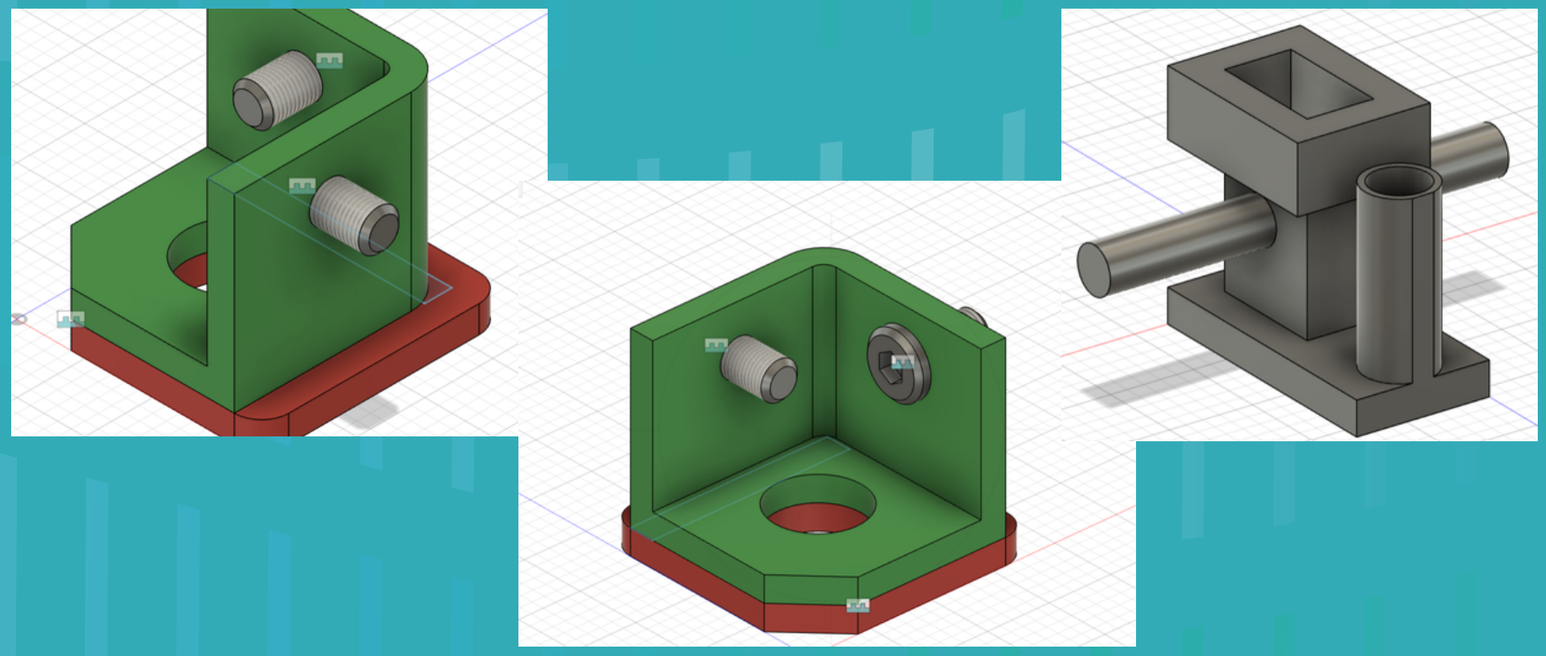
# Engineering PORTFOLIO

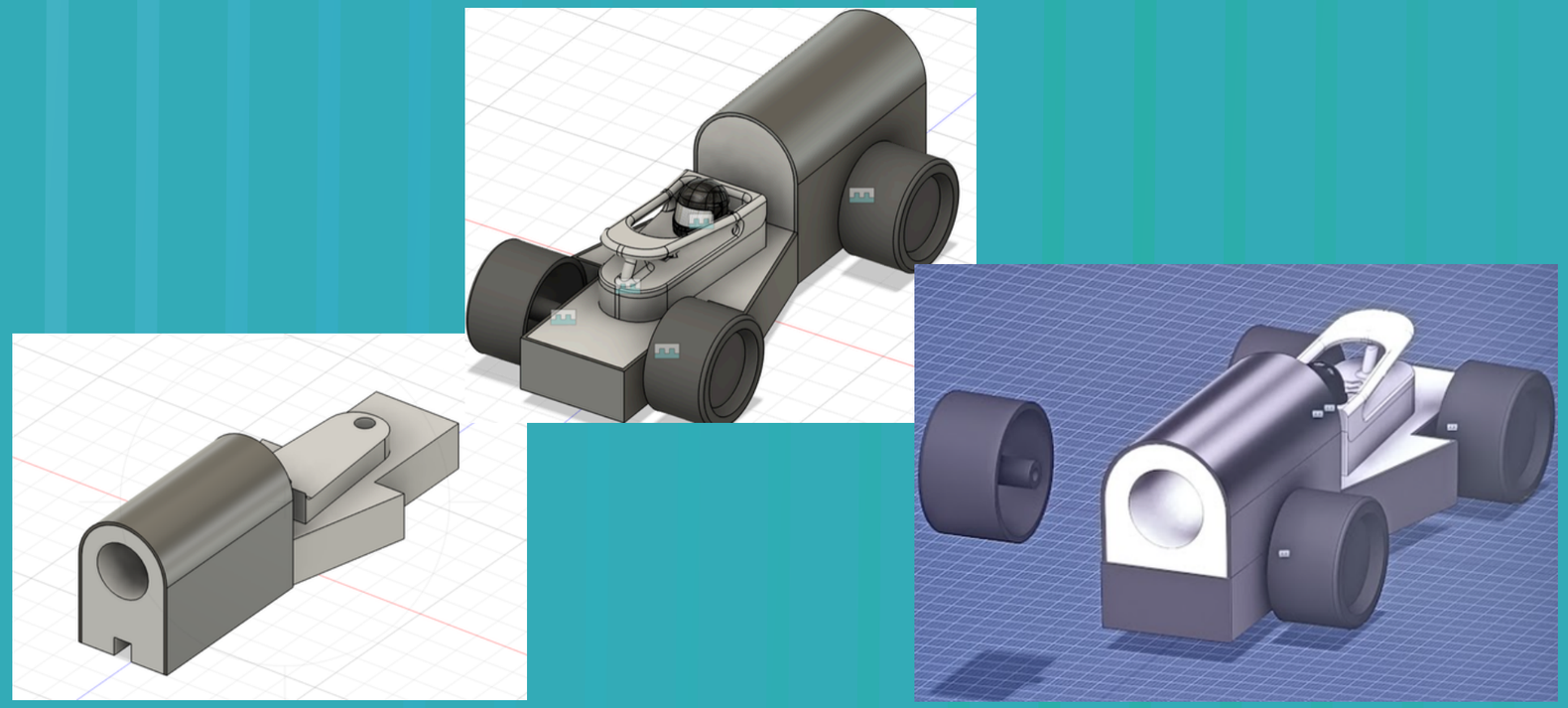


# CAR DEVELOPMENT

PNTHR RACING

We went on many websites, searched through ways on how to make something aerodynamical. Then we started learning how to model things like the car, we started it really early thankfully and this got us prepared for the summer. Before starting on the car, we watched videos of how others built theirs so we could follow their example. We began working on the computer, but to assemble the car, we had to build it one part at a time. After completing a section, we had to verify that it was constructed correctly and without errors; if there was a mistake, we would have to start all over again. And this actually happened after we finished our first design. We wanted to make sure it was printable but sadly it was not even exportable. So we started our second model and hoped it did not break again.



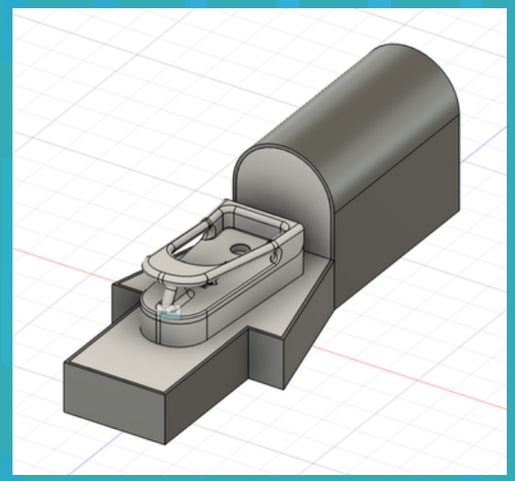


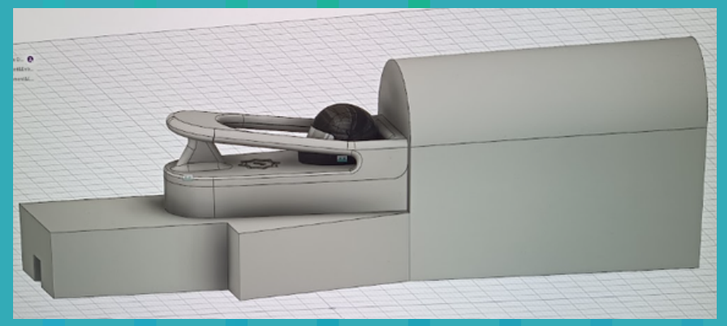
# CAR DESIGN

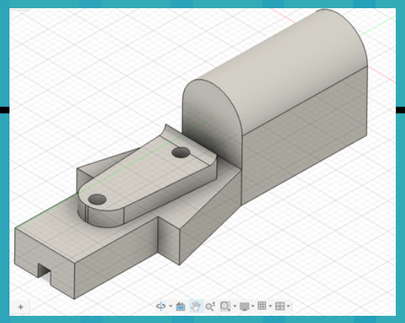
PNTHR RACING

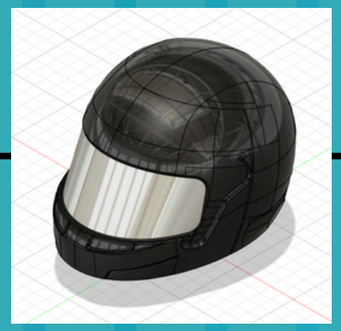
## The Main Parts

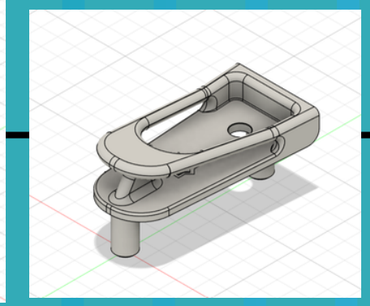
We started modeling the car, starting with the no-go-zone, halo, and helmet of the car. The no-go-zone, as its name suggests, is the part of the car you cannot touch, extrude, or cut while modeling. We started with a 1mm ring around the perimmeter of the car to not modify the no-go-zone for anything that we add to the model.

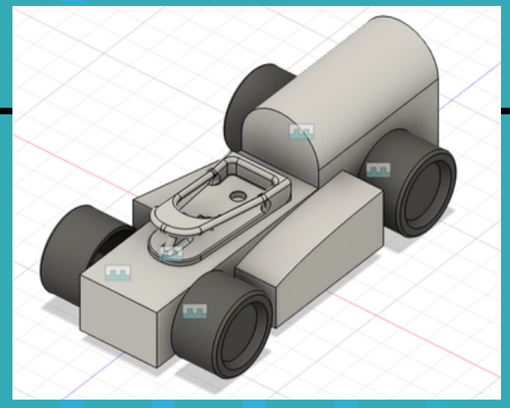


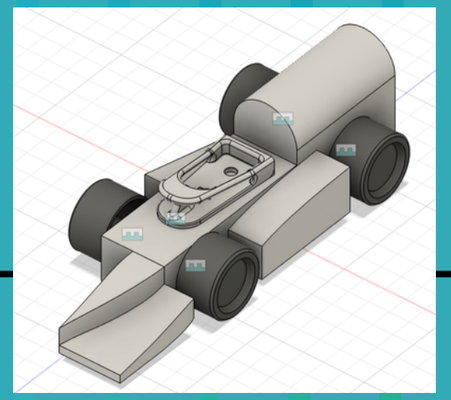


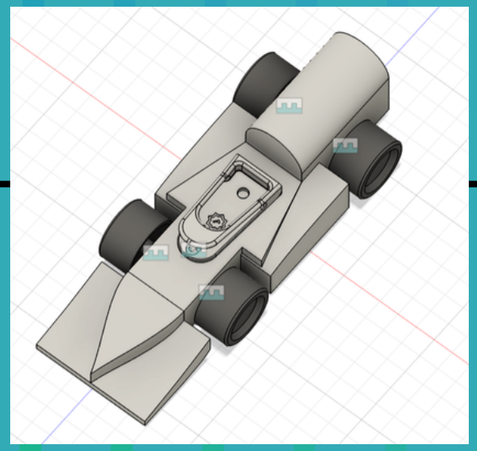












## Building development

The car slowly got bigger. We started with the sides of the car, putting what we have learned to use made some side chunks to make it have its bends and curves for an aerodynamical structure. The bends did take a while though, percisely pin-pointing an acurate measurement on the slope and making aerodynamical curves with the sketch tool did take us a quite large amount of time. This is because the net that we had built was pin-pointed incorrectly so we had to carefully move our way back from all of the progress we had already done. The fix did not leave a mark on the no-go-zone and that drops us of to the next step.

Then came the front of the car, making drawings on the cars face and extruding it outwards making a block at the front we can use for the nose. The nose started with sketches, mapping out where to cut using splines and where to extrude from meticulously pin-pointing where everything should go into place that does not cut the no-go-zone. To finish the base, we created a plane to mirror everything on the otherside without the two halfs inersecting eachother.

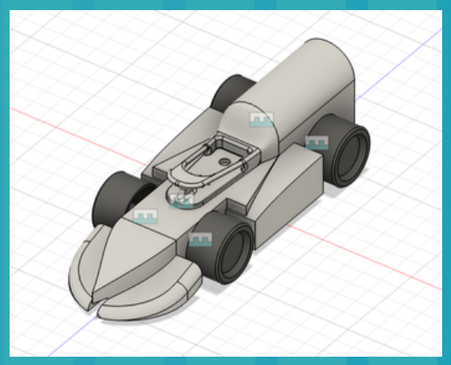
# CAR DESIGN

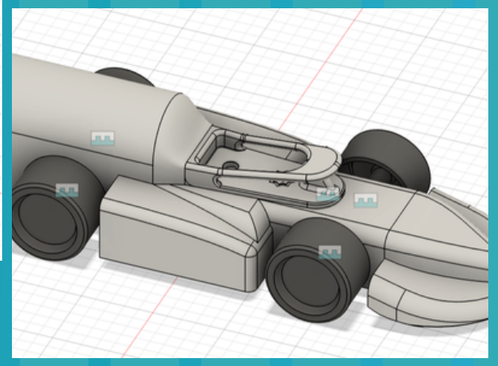
PNTHR RACING

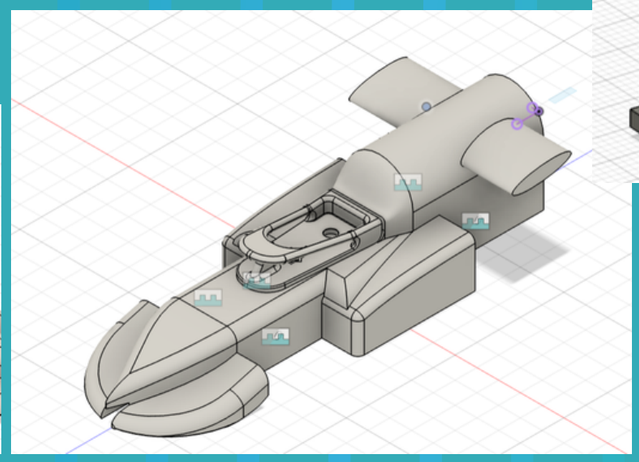
## The Back Wing

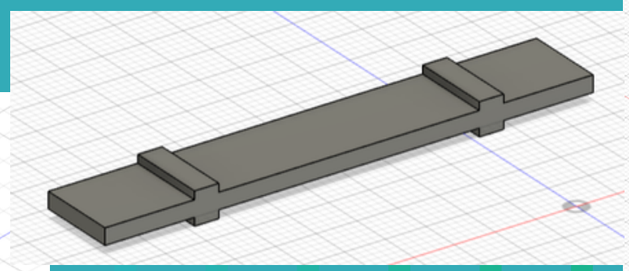
The back wings are made using a mix of sketches, elipsis, which are modifiable circles, and extrusions and our first design was one from the tutorial. It did have the aerodynamics for the racecar but we found out there were better ones out there so we went investigating and found a another design for the back wing. We used a mix of sketches, lines, and extrusions to make a new part for it to be inserted manually onto the cars back. A new part was needed because when we first gave in the design, we found out is was not possible to be lazer cutted out of the cube. We tried once more with the same outcome, however, we could print the piece. And that is what we did. This did occur after the main car model was made.

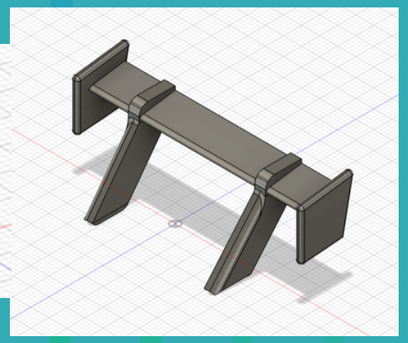
When we finshed with the pleasing model, we 3D printed it on an A1 Bambu Lab printer. Before printing, we used a set of supports that when used could print the back wing without it just falling over. Our first try was unsuccesful with a spagetti like print just moving the block around the place, but soon we got our back wing and it fit perfectly on our car. This test broke soon after so then we tried once more and got two perfectly intact backwings to fit on our car.





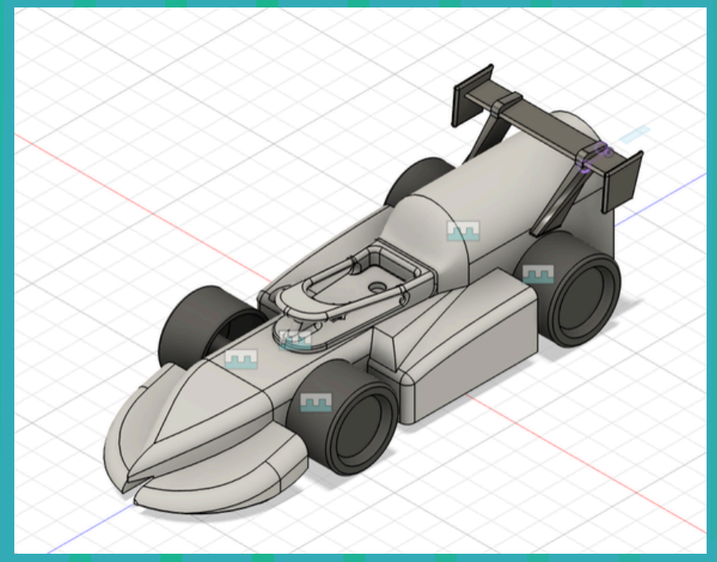






## Last Touches

We last worked on the fillets and chamfers. Fillets are the curves you give to vertices and edges to give it a streamlike apperance and Chamfers work the same way but instead make slopes. So we continued with adding the fillets and chamfers to the car, making carefull cuts to make it aerodynamical. Now came the opening at the back of the car and we used a loft. A loft connects two different sized faces together with a smooth wave to connect both sides. This is done by selecting two edges and points from both faces and connecting them to each other. This was our final design.



# Manufacturing

PNTHR RACING

All of the manufacturing occured first digitally. In Fusion 360, we simulated a milling process with a Denford 2600 Pro F1 machine, which is the main machine used when milling out the design for the teams cars. A block is placed on the machine with exact measurements to perfectly mill the block into the shape of the car's design chosen by each team. Then after making sure evrything was milled properly, the car was manufactured into its shape in a real Denford 2600 Pro F1.

After getting our milled cars, we sandpapered the car for it not to be dusty and for the paint to be as good as possible. The block is like a hard foam that has tiny holes where paint and water can seep through. So we decided to use a glace where it would be less dusty and able to be painted on without causing any holes and bubbles. We chose a blue/green turquoise and white representing our school's instinctive logo.

Then lastly were the stickers and adding the halo, helmet, wheels, and the back wing to the car. The stickers include the Stem Racing logo and our teams logo; the halo and the helmet pop right ont he no-go-zone of the car's model. Some halos' broke along this path including about two halos' snapped in half in less than two weeks. Lastly was carefully slidding on the back wing and we used hot glue to stick it on and for it to leave a small gap at the top of the back side of the car to make it more aerodynamical. And that lands on to the final design of our cars shown on the cover for this portfolio.

